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Week 4 practical

Tasks:

Step 1: Enable OSPF

1. On R1: router ospf 1 network 192.168.10.0 0.0.0.255 area 0 network 192.168.20.0 0.0.0.255 area 0 network 192.168.30.0 0.0.0.255 area 0 network [R1-R-EDGE link] [wildcard] area 0
2. On R-EDGE: router ospf 1 network [R1-R-EDGE link] [wildcard] area 0 network [R-EDGE's LAN] [wildcard] area 0

Step 2: Verify

1. show ip ospf neighbor
2. show ip route ospf
3. Ping from a VLAN PC to a device behind R-EDGE to confirm end-to-end connectivity.

The screenshot displays the Cisco Packet Tracer interface. On the left, the 'Logical' tab shows a network topology with a central 2960 24TT Switch1 connected to a 2911 R-EDGE router. Switch1 is also connected to two 2960 24TT Switches (Switch2 and Switch3). Switch1 is connected to three PCs (PC-SALES, PC-HR, and PC-IT). Switch2 is connected to PC-IT. Switch3 is connected to PC-IT. The R-EDGE router is connected to the Internet (represented by a cloud icon). The 'Physical' tab shows the same topology with physical connections.

On the right, the 'CLI' tab shows the configuration for R1. The configuration includes enabling OSPF, setting the router ID to 1.1.1.1, and configuring the OSPF network statements for the three areas (0.0.0.0/24, 0.0.0.0/24, and 0.0.0.0/24).

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IOS Command Line Interface

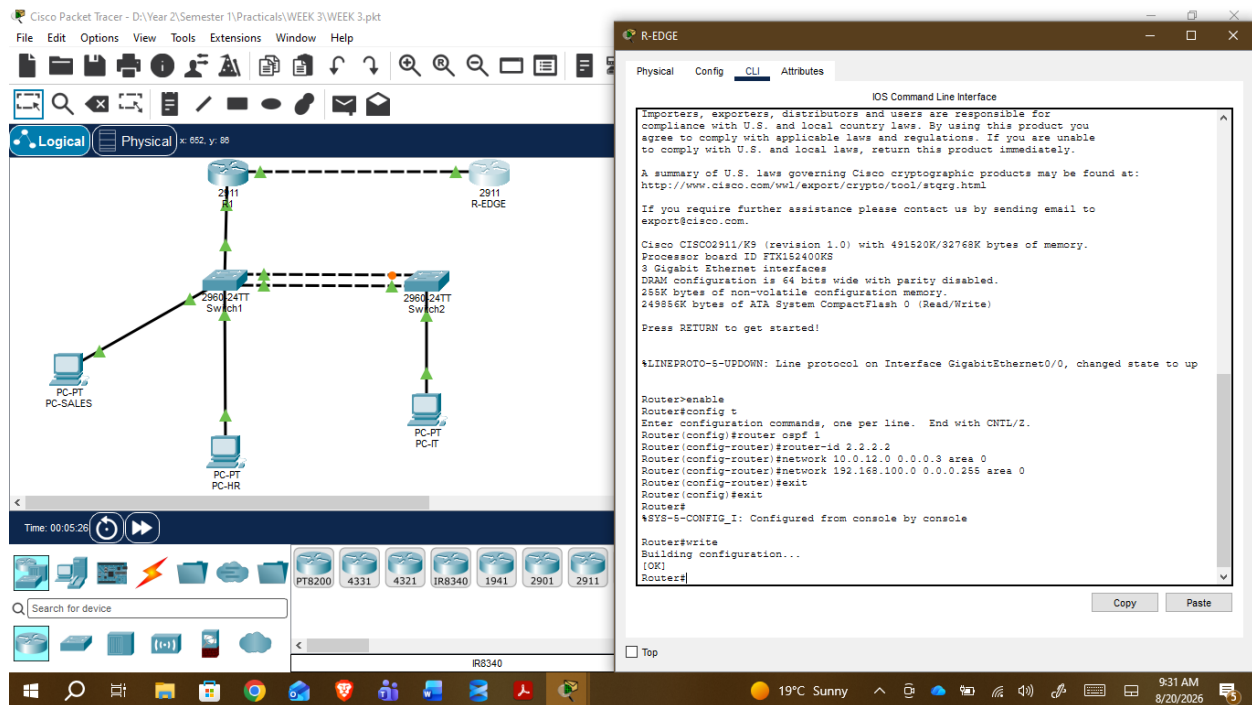
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 10.0.12.0 0.0.0.3 area 0
Router(config-router)#exit
Router#
$SYS-S-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#
```



Reflection Questions (answer these in your submission):

1. Compare troubleshooting a static-route failure (Week 3) to an OSPF neighbor failure (this week) — which was easier to diagnose, and why?

Troubleshooting an OSPF neighbor failure is easier to diagnose than a static-route failure.

Static route (Week 3): It fails silently. show ip route will just not show the route, or show it with a wrong next-hop. There is no protocol telling you why. You have to manually check every router: Did I type the next-hop IP correctly? Is the next-hop reachable? Is the exit interface down? It's time-consuming and error-prone, especially as the network grows.

OSPF (Week 4): OSPF has built-in diagnostic commands. show ip ospf neighbor immediately tells you if adjacency failed. show ip ospf interface shows if you are in the wrong area, and debug ip ospf hello shows mismatched timers or authentication. Once the neighbor is FULL, routes are learned automatically, so human typo errors on destination networks are eliminated.

Therefore OSPF is easier to troubleshoot because it gives you protocol states and self-heals - you fix the neighbor relationship once, and all networks are exchanged.